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DEPARTMENT OF CHEMICAL ENGINEERING

PREPARATION OF FRUIT LEATHER FROM WATERMELON RIND

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ABSTRACT

This study explores the valorization of discarded watermelon rind into a value-added fruit leather product. A processing route consisting of washing, size reduction, blending, formulation, heating, de-aeration, spreading, and controlled drying was developed. The effects of processing conditions were evaluated through measurements of soluble solids (°Brix), product yield, and sensory characteristics. The resulting process demonstrates the feasibility of converting watermelon rind waste into an edible product while providing an opportunity to optimize processing conditions and reduce organic waste.

INTRODUCTION

Watermelon rind is a significant portion of the fruit that is commonly discarded as organic waste, despite containing useful components such as dietary fiber, carbohydrates, and other bioactive compounds. This study explores the valorization of watermelon rind by converting it into fruit leather, creating a value-added food product while reducing fruit waste.

OBJECTIVES

- To develop a process for converting watermelon rind into fruit leather.
- To determine the optimum condition for the preparation of fruit leather.
- To evaluate the properties of prepared fruit leather such as yield percent, °Brix and sensory properties.
- To assess the environmental potential of watermelon rind valorization through food-waste reduction and preparing value-added product.

PROCESS FLOW CHART

RAW MATERIAL
↓
WASHING
↓
CUTTING
↓
BLENDING
↓
HEATING (at 60°C for 5 min)
↓
MIXING
↓
DE-AERATION (at 90°C for 5 min)
↓
DRYING (at 80°C for 5 hours)
↓
FRUIT LEATHER

Figure 1. Step by Step Diagram of Preparation of Fruit Leather

Table 1. Yield Percent and °Brix of Prepared Fruit Leather at Different Drying Time and Temperature

Sr. No.	Sample No.	Drying Temperature	Drying Time	Yield Percent	°Brix
1	Sample 1	60 °C	8 hr	56.41	7.5
2	Sample 2	80 °C	5 hr	57.53	8.7
3	Sample 3	100 °C	4 hr	50.069	9

CONCLUSION

The fruit leather was prepared by utilizing watermelon rind as raw materials, demonstrating its potential as a value-added product from fruit waste. Among the different drying conditions, the prepared fruit leather dried at 80°C for 5 hours is the optimum condition because it provided the best overall performance, achieving the highest yield (57%) and a relatively high °Brix value (8.7).